

## PinProbe A1

# Box Control Program Operation Guide

Device Connection / Common Controls / Exception Handling

Customer Integration Material

SCPI / UART

2026-07-28

This document is intended for customer-side host integration, device control, and maintenance reference.



# PinProbe A1 Box Control Program Operation Guide

Customer Release Material



2026-07-28  
Operation Guide  
PinProbe A1

Applicable program: PinProbeA1 Box Control Program.exe Document version: 2026-07-28 Applicable device: PinProbe A1 box control device

## 1. Preparation Before Use

1. Connect the PinProbe A1 device to the computer serial port or a USB-to-serial module.
2. Confirm the device is powered on and the serial driver is installed correctly.
3. Double-click PinProbeA1 Box Control Program.exe .
4. Default communication parameter is 115200 bps ; baud rate does not need to be changed.

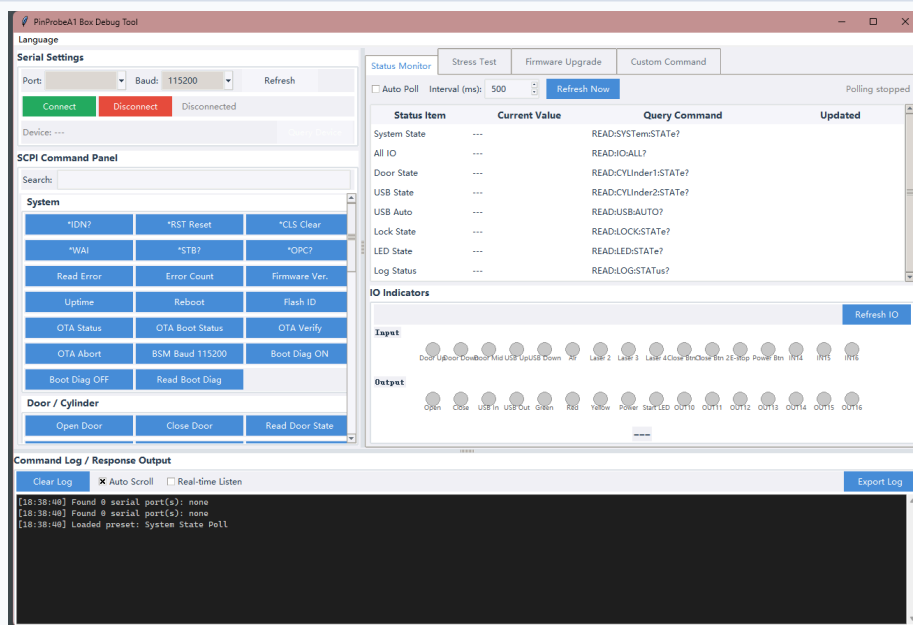
The program is portable and can run directly from the delivery package. If Windows shows a security prompt, choose "Run anyway".

## 2. Connect the Device

1. In the upper-left "Serial Settings" area, click "Refresh".
2. Select the device serial port from the "Port" drop-down list, such as COM3 , COM5 .
3. Click "Connect".
4. After connection succeeds, the status bar shows "Connected", and the program automatically queries device information.
5. Click "Query Device" to confirm the device returns GTS,PINPROBEA1,... .

If connection fails, check:

| Issue                 | Handling Method                                                                 |
|-----------------------|---------------------------------------------------------------------------------|
| Serial port not found | Check USB-to-serial driver, cable, and device power supply                      |
| Serial port occupied  | Close other serial tools and reconnect                                          |
| No response           | Confirm device power and selected port; replug serial cable and click "Refresh" |



Box control program main interface

## 3. Common Controls

The left "SCPI Command Panel" contains common buttons. Daily use should rely on buttons, not manual commands.

### 3.1 Door / Cylinder

| Button                    | Function                                                                  |
|---------------------------|---------------------------------------------------------------------------|
| "Open Door"               | Open the box door                                                         |
| "Close Door"              | Close the box door                                                        |
| "Read Door State"         | Query door position/action state                                          |
| "USB Insert"              | Insert USB manually (send <code>CONFigure:CYLInder2 CLOSE</code> )        |
| "USB Unplug"              | Unplug/retract USB manually (send <code>CONFigure:CYLInder2 OPEN</code> ) |
| "Read USB State"          | Query USB insert/retract state                                            |
| "Read USB Auto (Factory)" | Query factory USB auto setting                                            |

### 3.2 USB Automatic Sequence

USB auto plug/unplug is factory-set by hardware. Devices with a USB cylinder use `ON` ; others use `OFF` .

Recommended use:

1. After connection, click "Read USB Auto (Factory)" to confirm.
2. With a USB cylinder, keep `ON` ; USB inserts before close and retracts after open.
3. This setting **must not be modified**; use "USB Insert" / "USB Unplug" for debug.

### 3.3 Door Lock and Indicator Light

| Button                       | Function                             |
|------------------------------|--------------------------------------|
| "Unlock"                     | Unlock after power-on; allow actions |
| "Lock"                       | Lock device; stop normal actions     |
| "Read Lock State"            | Query locked/unlocked state          |
| "Green / Red / Yellow / Off" | Manually control indicator light     |
| "Read LED State"             | Query indicator-light state          |

## 4. Status Monitoring

The right "Status Monitor" tab shows current device state.

Functions:

| Function        | Description                                                 |
|-----------------|-------------------------------------------------------------|
| "Auto Poll"     | Periodically refresh system, door, USB, lock, LED, and logs |
| "Interval (ms)" | Polling interval; recommend <code>500 ms</code> +           |
| "Refresh Now"   | Refresh status once manually                                |
| "IO Indicators" | Quickly observe input/output states                         |

Key daily states:

| Status Item      | Common Return                                                         | Meaning                           |
|------------------|-----------------------------------------------------------------------|-----------------------------------|
| System State     | <code>LOCK</code> / <code>IDLE</code> / <code>READY</code>            | Locked, idle, or ready            |
| System State     | <code>RUNNING</code> / <code>COMPLETE</code> / <code>EMERGENCY</code> | Running, complete, or protected   |
| Door / USB State | <code>OPENING</code> / <code>CLOSING</code>                           | Action in progress                |
| Door / USB State | <code>OPENED</code> / <code>CLOSED</code> / <code>CYL ERR</code>      | At position or actuator exception |

## 5. Logs and Exception Handling

The lower log area shows commands, responses, and exceptions.

Operations:

| Function           | Description                                                      |
|--------------------|------------------------------------------------------------------|
| "Clear Log"        | Clear program-window content                                     |
| "Auto Scroll"      | Scroll to bottom when new logs appear                            |
| "Real-time Listen" | Receive live controller logs for integration/replay              |
| "Export Log"       | Save logs as <code>.log</code> or <code>.txt</code> for analysis |

When an exception occurs:

1. Stop auto test or continuous operation.
2. Check logs for `ERROR`, `ERR`, or `WARN`.
3. Click "Refresh Now" in "Status Monitor".
4. Read door, USB, lock, and system states.
5. Export logs for technical support.

Common prompts:

| Prompt                        | Meaning                   | Handling                                    |
|-------------------------------|---------------------------|---------------------------------------------|
| <code>ESTOP</code>            | E-stop triggered          | Check E-stop button; reset and unlock again |
| <code>LASER</code>            | Anti-pinch or obstruction | Clear obstruction; confirm safety and retry |
| <code>RS485_FAULT</code>      | IO communication fault    | Check connection/power; wait or restart     |
| <code>USB_INSERT_FAIL</code>  | USB insertion failed      | Check USB insert mechanism and fixture      |
| <code>USB_RETRACT_FAIL</code> | USB retraction failed     | Check USB retract mechanism and fixture     |
| <code>IO_WRITE_FAIL</code>    | Actuator output failed    | Check IO module, power, and cables          |

## 6. Custom Commands

Use "Custom Command" for manual SCPI during maintenance/integration.

Usage:

1. Enter a command, e.g. `*IDN?`.
2. Click "Send" or press Enter.
3. View the lower log response.
4. Click history to reuse.

Daily use needs no manual input. See the bundled SCPI Command Reference.

## 7. Stress Test

"Stress Test" loops commands for factory check, aging test, or issue replay.

Recommendations:

| Item             | Recommended Value                                 |
|------------------|---------------------------------------------------|
| Command interval | No lower than <code>500 ms</code>                 |
| Loop count       | Use finite count for site verification            |
| Timeout          | Set by action duration; reserve door-command time |

During stress test, some auto polling is paused to reduce serial conflicts. Re-enable "Auto Poll" after testing.

## 8. Firmware Upgrade

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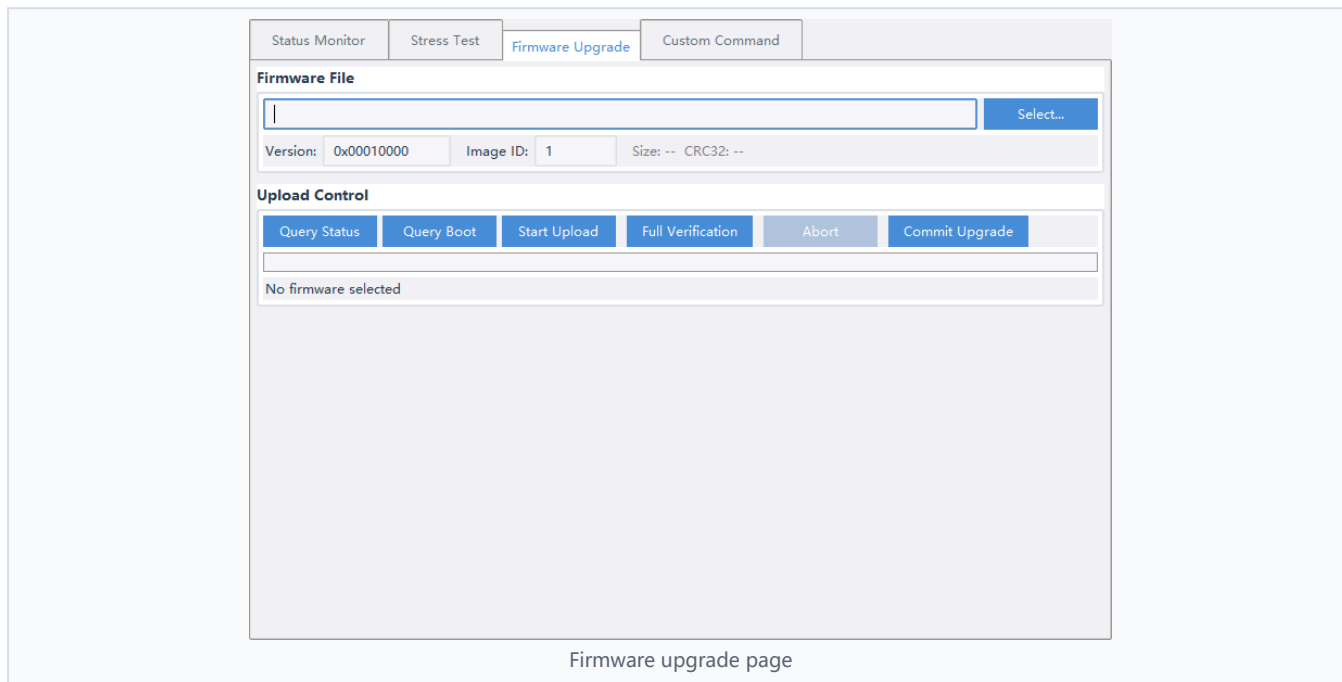
### 8. Firmware Upgrade

The "Firmware Upgrade" tab is for OTA upgrade. Customer sites should not upgrade without technical support guidance.

Basic procedure:

1. Click "Select..." and choose the `.bin` firmware file.
2. Confirm displayed file size and CRC32 information are normal.
3. Click "Full Verification"; the program completes upload, verification, commit, and post-reboot status check.
4. Wait for "Full verification passed" in logs before continuing.

"Start Upload" and "Commit Upgrade" are for technical support troubleshooting; customer sites should use "Full Verification" first.



Notes:

| Notice Item      | Description                                                     |
|------------------|-----------------------------------------------------------------|
| Firmware file    | Must select a PinProbe A1 App firmware <code>.bin</code> File   |
| Power            | Do not power off during upgrade                                 |
| Serial port      | Do not disconnect serial port during upgrade                    |
| Operation        | Do not perform door, USB, lock, or other actions during upgrade |
| Failure Handling | If upgrade fails, export logs and contact technical support     |

### 9. Delivery Package Files

| File                                                                              | Description              |
|-----------------------------------------------------------------------------------|--------------------------|
| <code>PinProbeA1 Box Control Program.exe</code>                                   | Box control host program |
| <code>PinProbeA1 Box Control Program Operation Guide Customer 20260728.pdf</code> | This operation guide     |
| <code>PinProbe A1 Box Control SCPI Command Reference 20260728.pdf</code>          | SCPI command reference   |

### 10. Recommended Operation Sequence

Run program -> refresh and select serial port -> click "Connect" -> click "Query Device" -> click "Read Lock State" and "Read System State" -> perform unlock, open door, close door, USB automatic or manual control as required; export logs if an exception occurs.